



BEng, BSc, MEng and MMath Degrees 2025–26

Open Assessment

Department Computer Science

Module DATA: Introduction to Data Science

Title Monday DATA Summative Assessment

Issued: 12 noon, Monday [27 April 2026]

Submission due: 12 noon, Monday [18 May 2026]

Feedback and Marks due to students: 12 noon, Tuesday [23 June 2026]

Word limit All tasks specify strict limits on the length of your answers; text exceeding the limit will not be marked. Please note that these limits apply to written responses only, and code is not subject to word limits.

This is an anonymous assessment. Do **not** include your name, student number or any other identifying information in the submitted material (including metadata).

All students should submit their answers through the submission point in the Assessment area of the VLE site by 12:00 noon, Monday, [18 May 2026]. An assessment submitted after this deadline will be marked initially as if it had been handed in on time, but the Board of Examiners will normally apply a lateness penalty. The first late submission will be marked if and only if no on-time submission has been made.

Your attention is drawn to the section about Academic Misconduct in your [Departmental Handbook](#).

Any queries you may have on this assessment should be posted on the Discussion Board on the Virtual Learning Environment (VLE) page for DATA: Introduction to Data Science in the appropriate discussion area. **No questions will be answered after Tuesday, 05/05/2026.**

Note on Academic Integrity

This is an open assessment, and you are permitted to refer to written and online materials to aid you in your answers. However, you must ensure that the work you submit is entirely your own, and for the whole time the assessment is live, you must not:

- communicate with other students on the topic of this assessment;
- seek advice or contribution from any other third party, including proofreaders, friends, or family members; or
- use Generative AI unless it is specifically permitted in the assessment specifications.

We expect, and trust, that all our students will seek to maintain the integrity of the assessment, and of their award, by ensuring that these instructions are strictly followed. Where evidence of academic misconduct is evident, this will be addressed in line with the [Academic Misconduct Policy](#) and, if proven, be penalised in line with the appropriate penalty table (see AM3.3 of the Academic Misconduct Policy).

Please note: The publication of your assessment submission online or elsewhere, or the sharing of these documents with others, is **NOT PERMITTED** until all students have completed the module assessment and reassessment (this will include any students with extensions to their reassessment date). We will notify you through the VLE at such time.

Background

Business Domain

In the rapidly evolving landscape of modern living, smart home technology has become a crucial component of energy management, sustainability, and comfort optimisation. Smart homes utilise advanced sensors, automation systems, and interconnected devices to monitor, record, and regulate household energy consumption with high precision. These systems generate large volumes of fine-grained data on household appliances, electricity generation, and environmental conditions. When effectively analysed, this data can provide valuable insights into energy efficiency, cost reduction, and environmental impact. However, much of the information remains in separate and unprocessed datasets, making it difficult to gain a comprehensive understanding of household energy performance and behaviour.

Imagine you are a data scientist working for a leading Smart Energy Research Consortium in the United Kingdom. Acknowledging the significant role of data-driven insights in promoting sustainable living, the Chief Data Officer (CDO) has established a Smart Environment Analytics Team, appointing you as its lead data scientist.

Aim: Your team is tasked with conducting a comprehensive data science project using a purchased smart home dataset. This dataset comprises minute-by-minute recordings collected over 350 consecutive days, encompassing both appliance-level power consumption (in kilowatts) and local weather conditions corresponding to the household's region.

The project's outcomes will be presented to key stakeholders, including energy analysts, environmental policymakers, and smart technology developers. The primary objective is to uncover insights through requirement-led exploration of data and patterns, thereby supporting the development of data-driven strategies that enhance energy efficiency and foster sustainable living.

Smart Home Data Overview

The dataset represents detailed sensor readings from a modern smart home equipped with an advanced smart metering system integrated with external weather monitoring data. Each record captures the state of energy consumption across various household devices, combined with concurrent weather observations, at a one-minute interval throughout a 350-day period. This high-resolution data offers a valuable opportunity to model energy consumption behaviour with respect to environmental influences.

In this dataset, consumption is recorded not just as total load but also as sub-metered, appliance-specific power draw for components like dishwasher and fridge, while the weather data includes critical variables such as temperature and humidity.

Tasks

You are required to design and build solutions based on the above introduction (page 3) and the data provided. There are three specific tasks: Task 1 – Task 3.

In particular,

- Task 1 (20%): Demonstrate a good understanding of statistical analysis, implementation, and the responsible use of AI tools in data driven decision making;
- Task 2 (40%): Conduct a domain analysis and design a database for the given scenario and data;
- Task 3 (40%): Design and execute a data analysis plan to derive actionable insights from the provided smart home data, addressing the business domain and requirements, and answering the analytical questions supplied.

A detailed description of each task is provided below.

Your findings and code should be presented as a Jupyter notebook and include the tasks

described below. The dataset (present in the file:HomeC.csv) and a Jupyter Notebook template are available on VLE (VLE → DATA → Assessment → Assessment 2025-26 resources). Further formatting details and word limit per task are given below.

Task 1: Technique Explanation and Responsible Use of AI tools (20 Marks)

Question 1.1 (5 Marks): Outlier Detection Using the IQR Method: Focus on the column with the heading ***use [kW]*** in the provided dataset, which contains unsorted electricity usage values and may include missing or corrupted entries resulting from transmission errors. **In bullet point form**, clearly and systematically describe the complete step by step statistical procedure required to identify the lower and upper outlier boundaries for this data using the Interquartile Range IQR method. Your description should begin with any necessary data preparation steps and proceed through to the definition of the outlier thresholds. No numerical calculations are required. However, your answer must demonstrate a clear, logically ordered, and statistically sound methodology, explicitly referencing all key stages involved in applying the IQR method. (100 words max)

Question 1.2 (5 Marks): Python automation: Building on Question 1.1, develop a Python script that performs automated outlier detection using the Interquartile Range IQR method for the dataset column with the heading ***use [kW]***. Your script should begin by correctly loading the dataset into the Python environment and appropriately handling any missing or corrupted values prior to analysis. Using correct Python syntax, compute the relevant statistical measures required for the IQR method and determine the lower and upper outlier boundaries.

Your answer must include the complete Python script, the printed values of the median, the lower outlier boundary, and the upper outlier boundary, as well as a clearly labelled box plot that visualises the data distribution that shows the detected outliers. (100 words max excluding Python script)

Question 1.3 (10 Marks): Responsible Use of Large Language Models and Analytical Validation (200 words max)

Scenario: A team member uses a large language model to analyse the dataset column with heading **use [kW]** and provides you with the following summary statistics:

Using the Interquartile Range (IQR) method on the use [kW] column, the outliers are defined as any energy usage values falling outside the calculated lower and upper bounds.

Here is the breakdown of the outlier analysis:

Lower Bound: -0.536 kW

Upper Bound: 1.874 kW

Outlier Definition: Any value smaller than -0.536 kW and greater than 1.874 kW is considered an outlier.

Summary Statistics for Outliers:

Total Count: 32,211 data points were identified as outliers.

Percentage: Approximately 6.39% of the recorded data are outliers.

Range: These high-usage outliers range from 1.874 kW up to a maximum of 14.715 kW.

These outliers represent periods of unusually high energy consumption compared to the typical usage patterns in the dataset.

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(a) Is the provided summary accurate? If not, clearly identify the issue.

(b) Briefly describe the specific process you used to independently verify these results.

(c) Identify one good practice that should be followed to ensure the responsible use of large language models in data analysis.

Task 2: Domain Analysis and Database Design (40 Marks)

Question 2.1: Conduct a literature review of the domain and use the knowledge obtained to list the entities/classes in the dataset provided as well as the attributes of each of the entities/classes. Discuss the rationale used and support the choices you have made as well as the level of resolution you have chosen. Do this in 350 words maximum [8 Marks].

Question 2.2: Using the entity/class list you created in Question 2.1, choose one of the methodologies used to represent domain knowledge. Use the methodology to create a pictorial representation of the domain including the links between the entities/classes in the dataset [6 Marks].

Question 2.3: Using the entity/class list you created in Question 2.1, create GUIs to support intelligent data entry and recovery from the database. Using up to 100 words, discuss how your approach is providing intelligence to the data retrieval process [6 Marks].

Question 2.4: Using the chosen methodology from Question 2.2 and 2.3 above, create a database of the domain above while respecting the relational links between the classes. In python code, give 5 examples of sql commands to interrogate the database and show the results obtained. In 200 words maximum, justify why they respect the constraints of a relational database [12 marks].

Question 2.5: In 500 words maximum, discuss and demonstrate how the concepts above have contributed to the way you extract data for further Machine Learning tasks [8 marks].

Task 3: Data Analysis and Insight Generation (40 Marks)

Question 3.1: Descriptive Statistics and Correlation (10 Marks):

a) Over the 350-day period, analyze the hourly average power usage for Furnace 1 [kW]: (5 Marks)

- 1- Find the hour of the day with the highest average usage.
- 2- Provide a possible reason for this pattern based on the function of a furnace. (70 words max)

b) Compute the correlation between Fridge [kW] and temperature over the entire dataset. (5 Marks)

- 1- Is the correlation positive, negative, or negligible?
- 2- What does this imply about the relationship between outdoor temperature and fridge energy usage? (80 words max)

Question 3.2: Supervised Learning (10 Marks):

a) Write a script to fit a linear regression model where the target (Y) is the 'Home office [kW]' consumption and the predictor (X) is the 'Living room [kW]' consumption (submit the script via Jupyter Notebook and discuss the results).

1. Fit the model.

2. Print the resulting R^2 score of the fitted model (using the same data for training and scoring for simplicity). Assume the required data reshaping has already been performed. (5 Marks)

b) Based on the names of the two variables used ('Home office [kW]' and 'Living room [kW]'), if the R^2 score is very close to zero, what conclusion can be drawn about the potential collinearity or independence of these two specific loads in a typical residential setting? (100 words max) (5 Marks)

Question 3.3: Classification (10 Marks)

Build a binary classification model to predict whether the Microwave [kW] is currently ON or OFF. Define ON as any reading greater than 0.1 kW and OFF otherwise.

a) Write the code to create a new binary target column named 'Microwave_State' where the value is 1 (ON) if Microwave [kW] > 0.1, and 0 (OFF) otherwise. Then, split the data (train_test_split) into X_train, X_test, Y_train, Y_test, using 'pressure' and 'humidity' as your feature columns (X). Use test_size=0.3 and random_state=42. (5 Marks)

b) Using the split data from part (a), fit a Logistic Regression model. After predicting on the test set (Y_pred), calculate and print the accuracy score. Based on the typical class imbalance in appliance use (Microwave is OFF most of the time), explain why the accuracy score might be a misleading metric for this specific problem. (80 words max) (5 Marks)

Question 3.4: Unsupervised Learning (10 Marks)

Use K-Means Clustering (k=3) to identify three distinct seasonal usage patterns based on two features: 'temperature' and 'House overall [kW]'.

a) Why is it necessary to apply a scaling technique (like Standardization or Normalization) to these two columns? Explain why the difference in typical scale (temperature might range from 0 to 30; House overall [kW] might range from 1 to 10) will distort the Euclidean distance calculation if scaling is omitted. (130 words max) (3 Marks)

b) Apply KMeans clustering with k=3 using the two standardized features. Then:

1- Show the resulting cluster centers (convert them back to original units).

2- Plot the scatter plot of the original features (temperature vs house usage), colored by cluster labels. (5 Marks)

c) Interpret the cluster centers. What does each cluster likely represent in terms of seasonal usage patterns (e.g., winter, summer, etc.)? (100 words max) (2 Marks)

Deliverables

You should submit the following to the submission point on VLE:

1. the SQLite database produced in Task 2;
2. the completed Jupyter notebook (both .ipynb and HTML files) containing solutions for all the tasks (figures or diagrams that are included in your answers can be included in the Jupyter notebook.).

END OF PAPER